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ALMA et al.(10) **Pub. No.: US 2025/0283346 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **ROOFTOP TENT ANNEX SYSTEM**(30) **Foreign Application Priority Data**(71) Applicant: **Thule Sweden AB**, Hillerstorp (SE)

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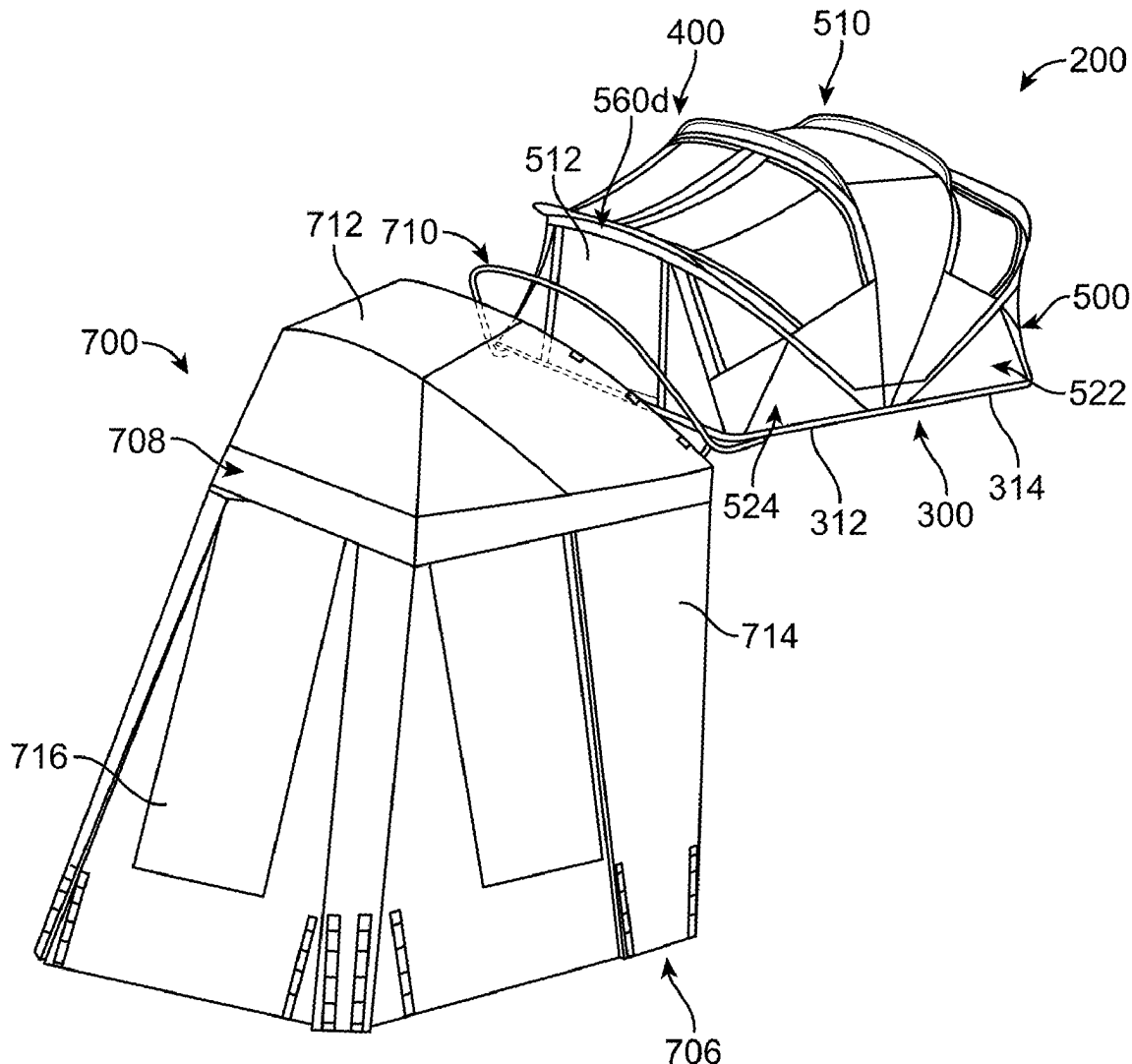
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(57) **ABSTRACT**

A tent system includes a base configured to mount to a vehicle, a base canopy, a frame system coupled to the base and configured to support the base canopy, and an annex. The annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. The annex is removably coupled to the base and the base canopy.



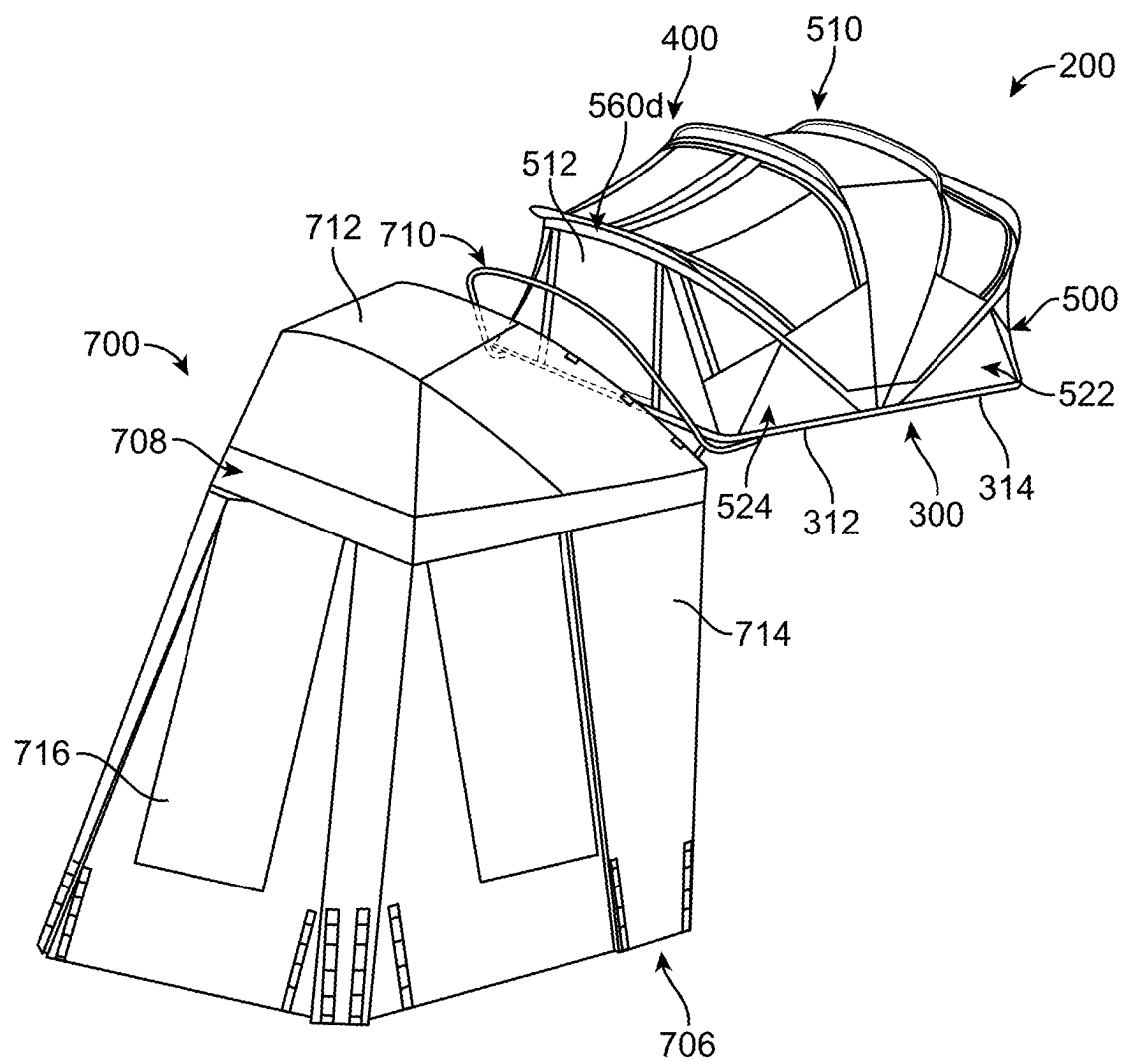


FIG. 1

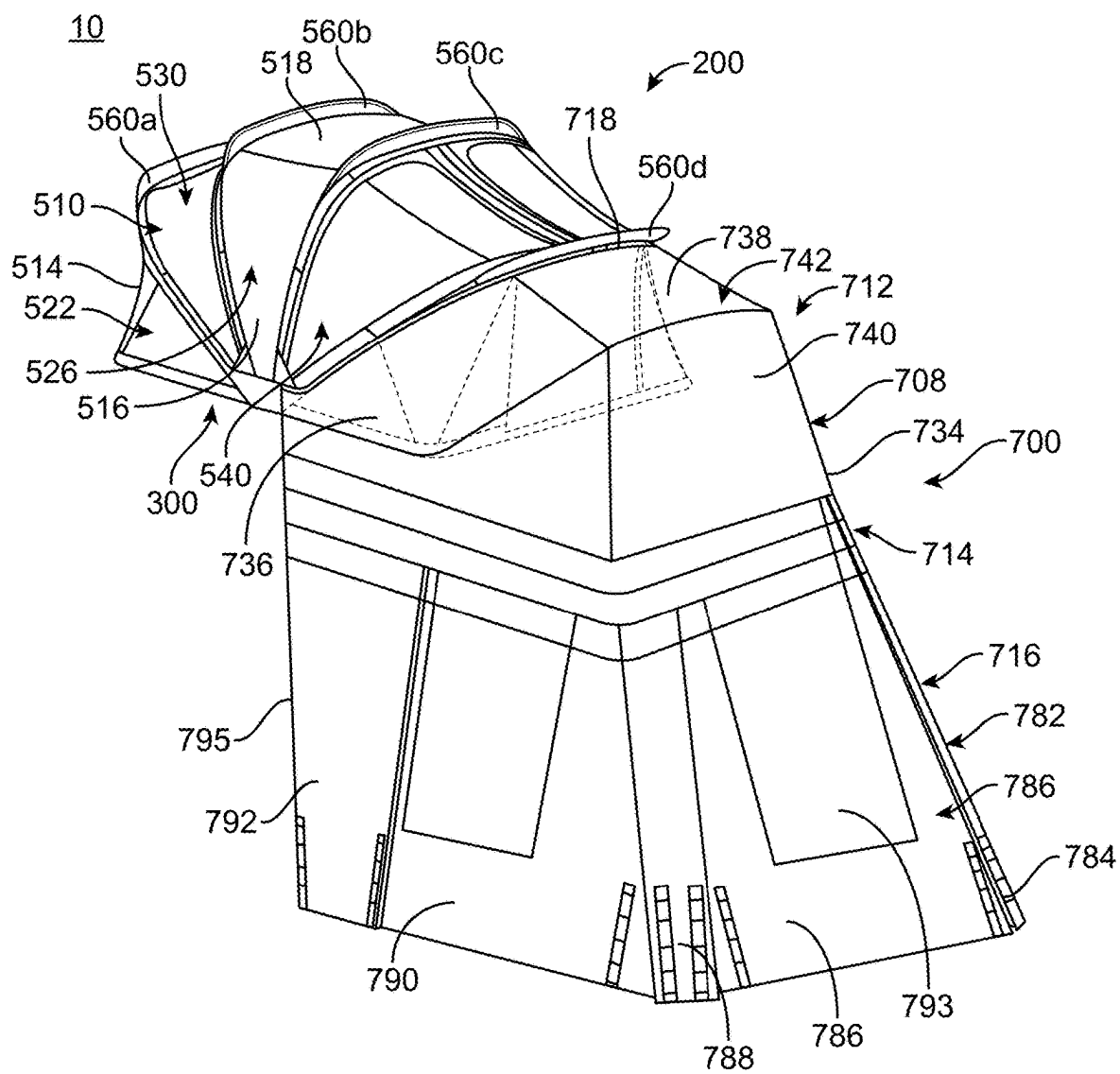


FIG. 2

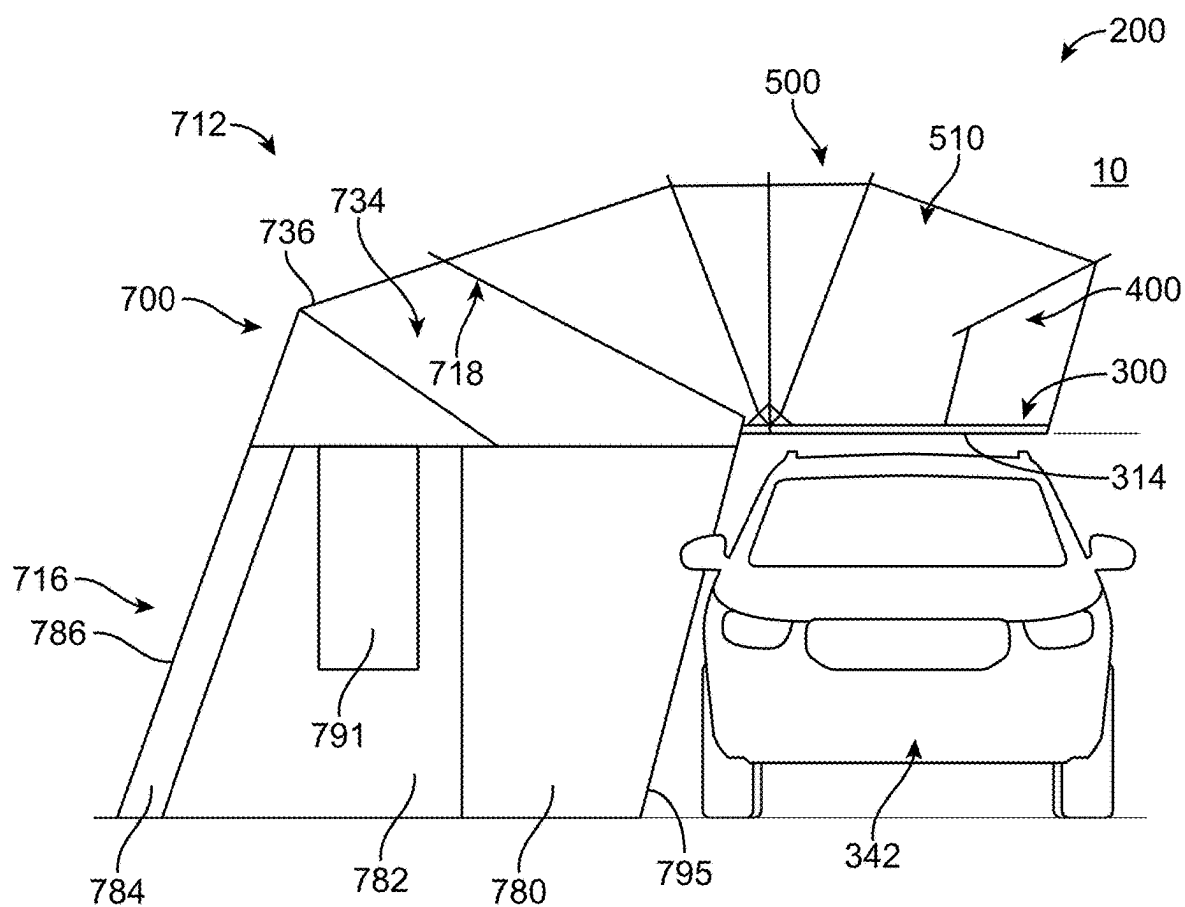


FIG. 3

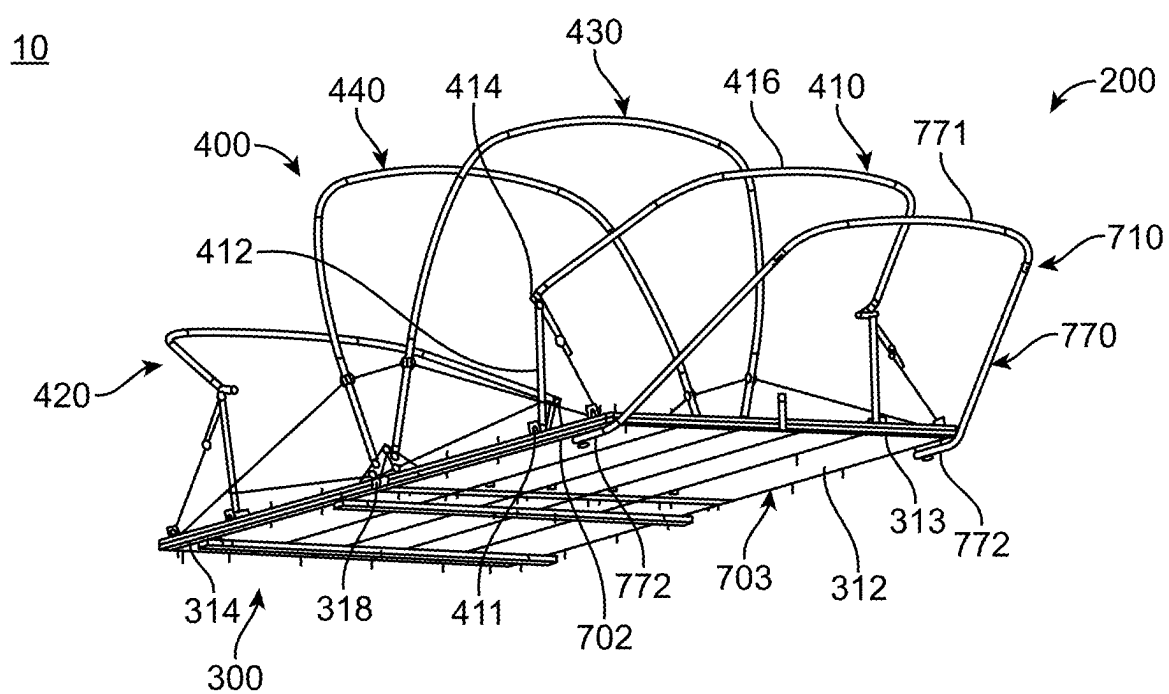


FIG. 4

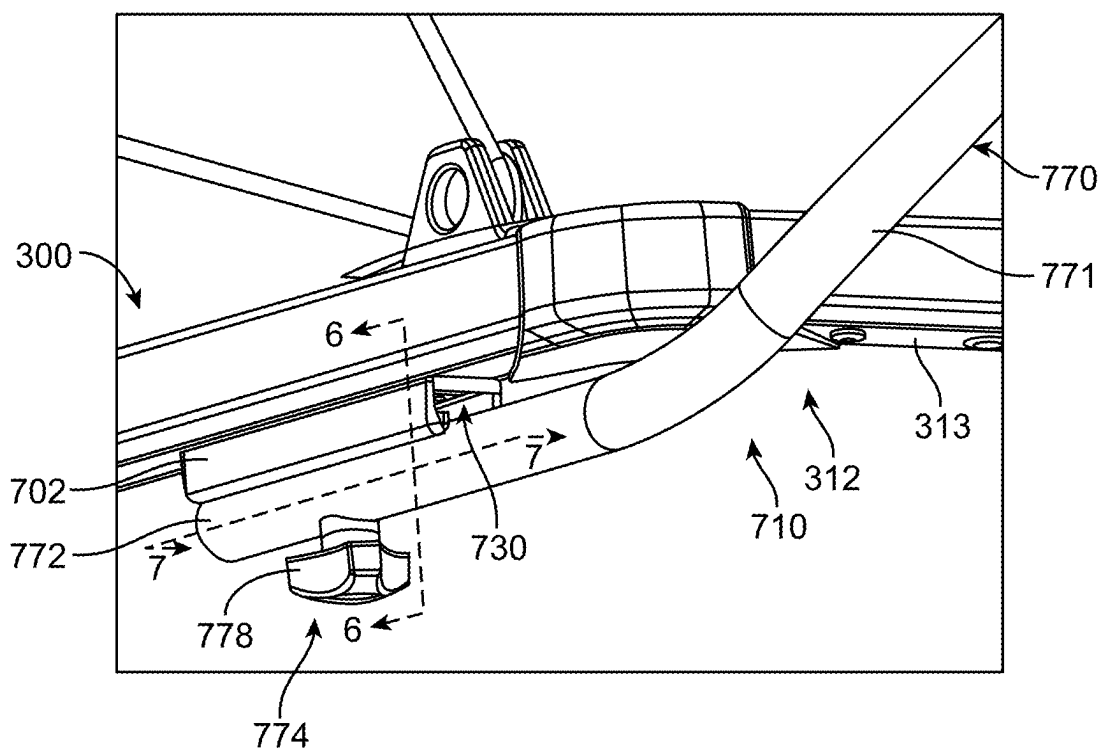


FIG. 5

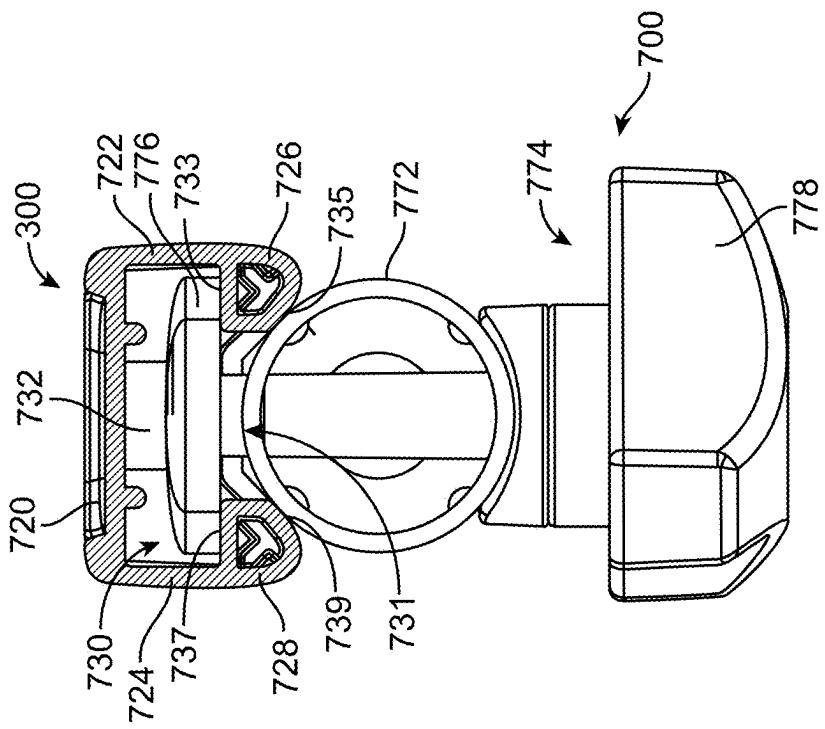


FIG. 6

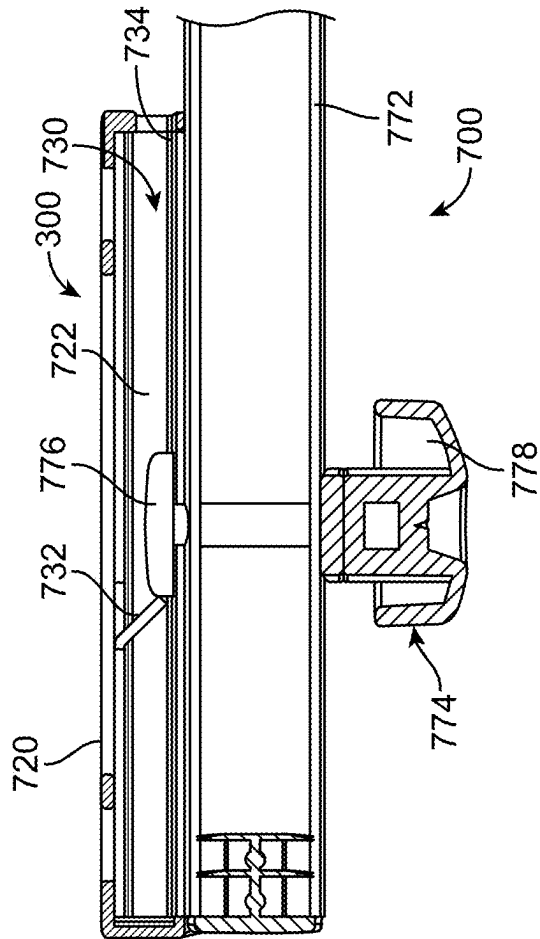


FIG. 7

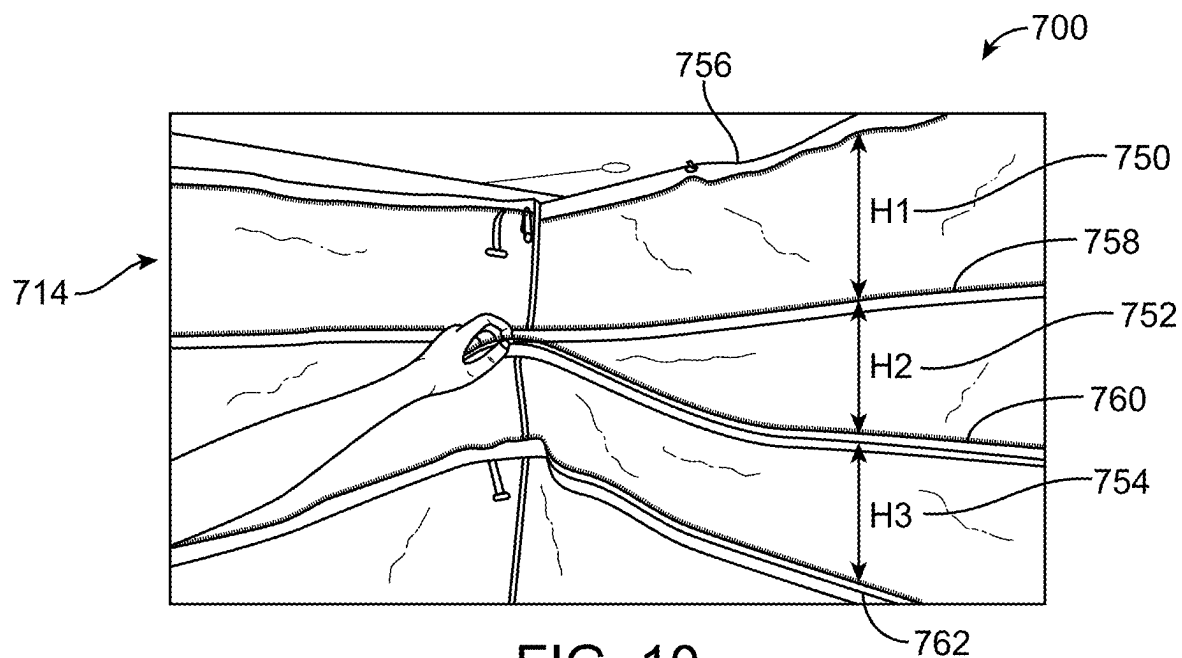


FIG. 10

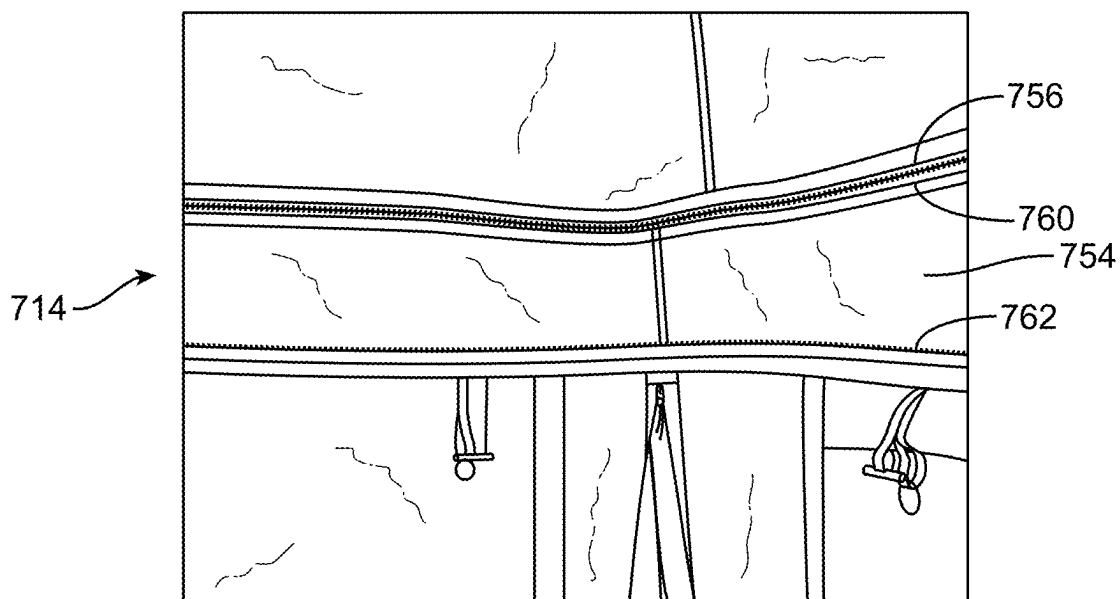


FIG. 11

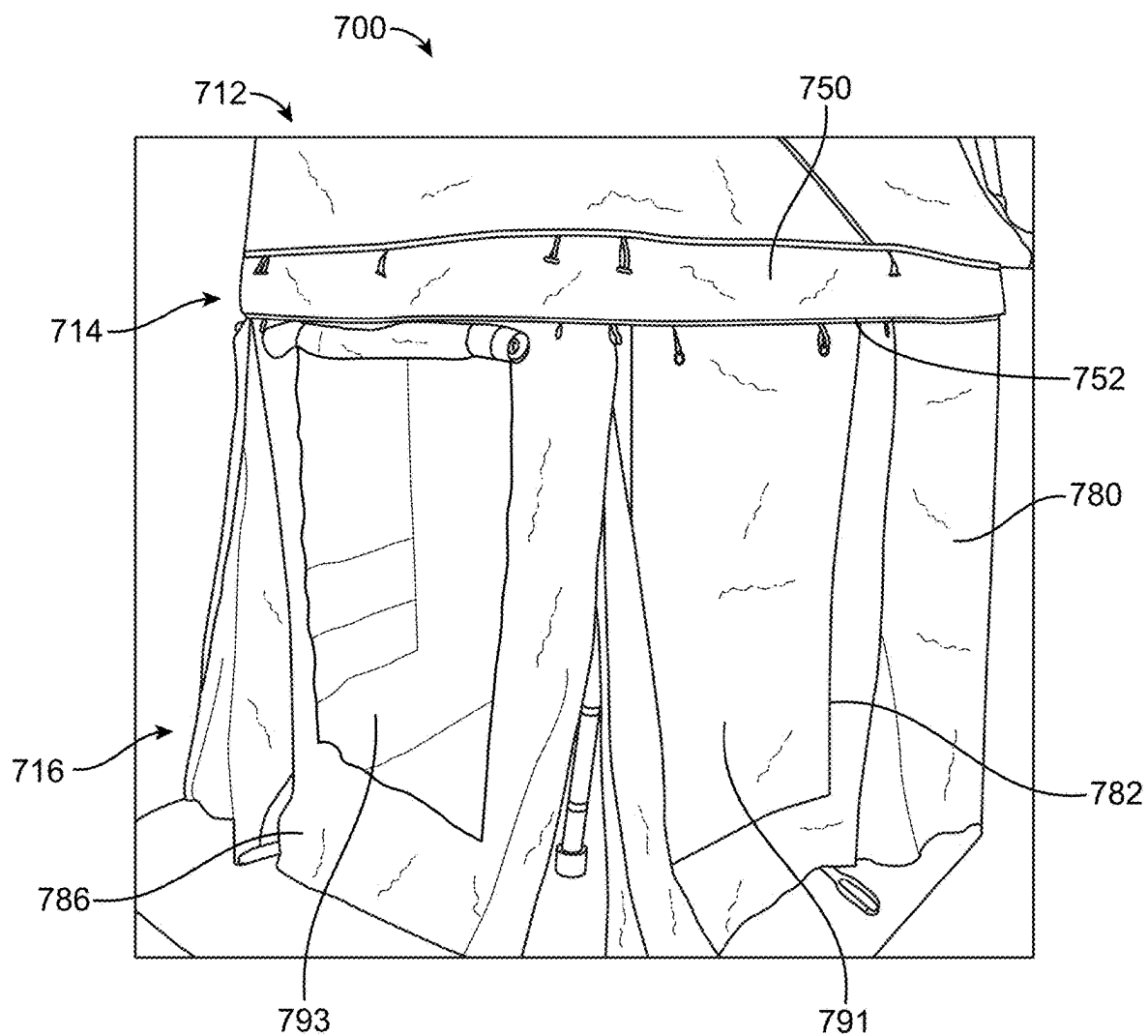


FIG. 12

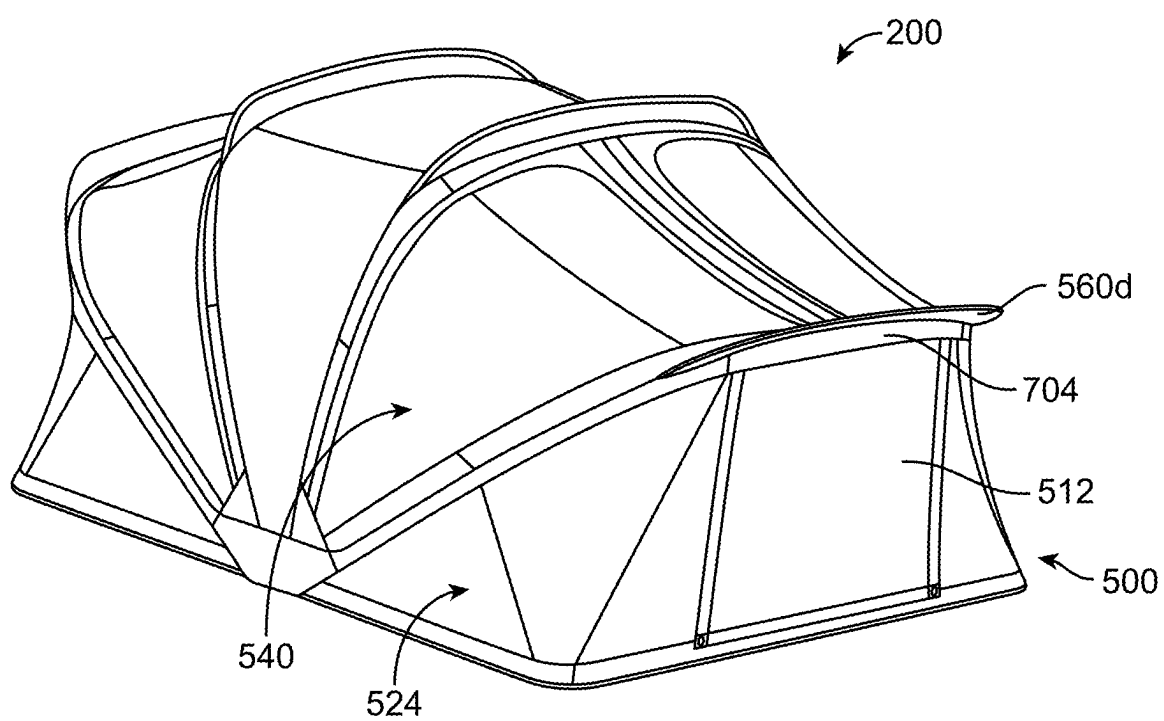


FIG. 13

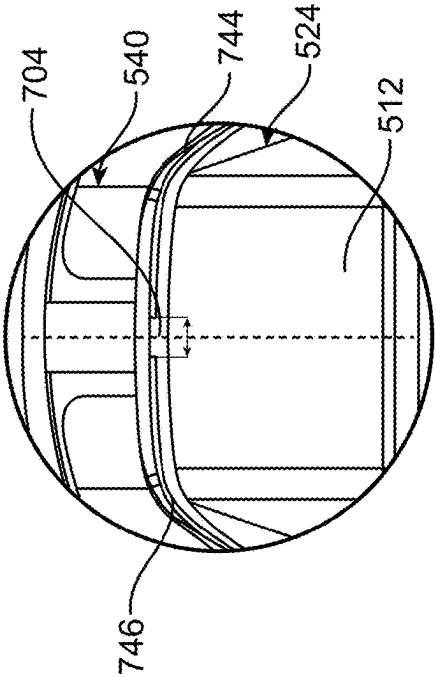


FIG. 14B

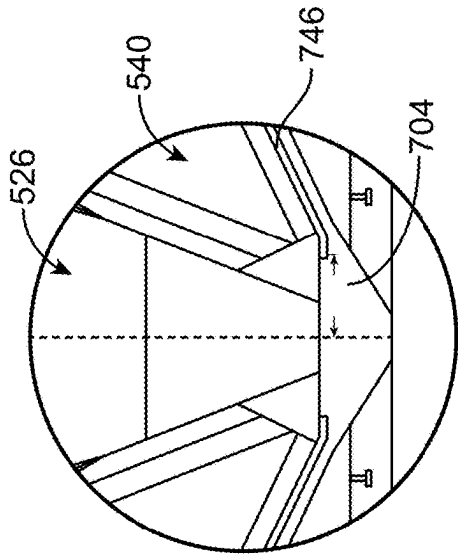


FIG. 14C

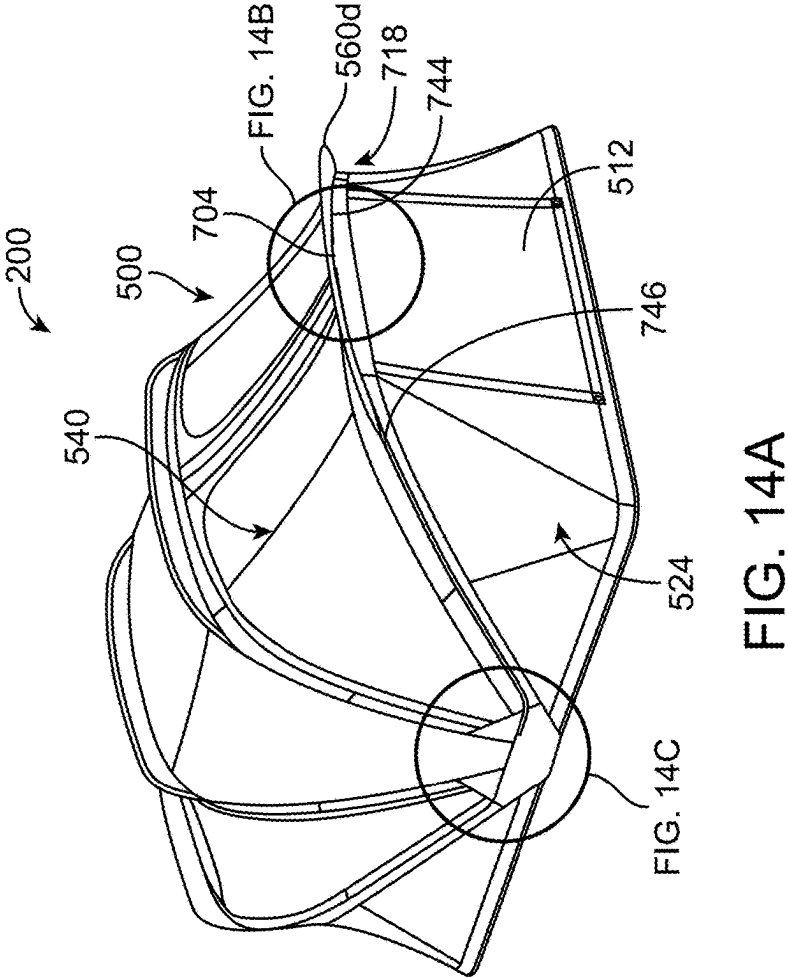


FIG. 14A

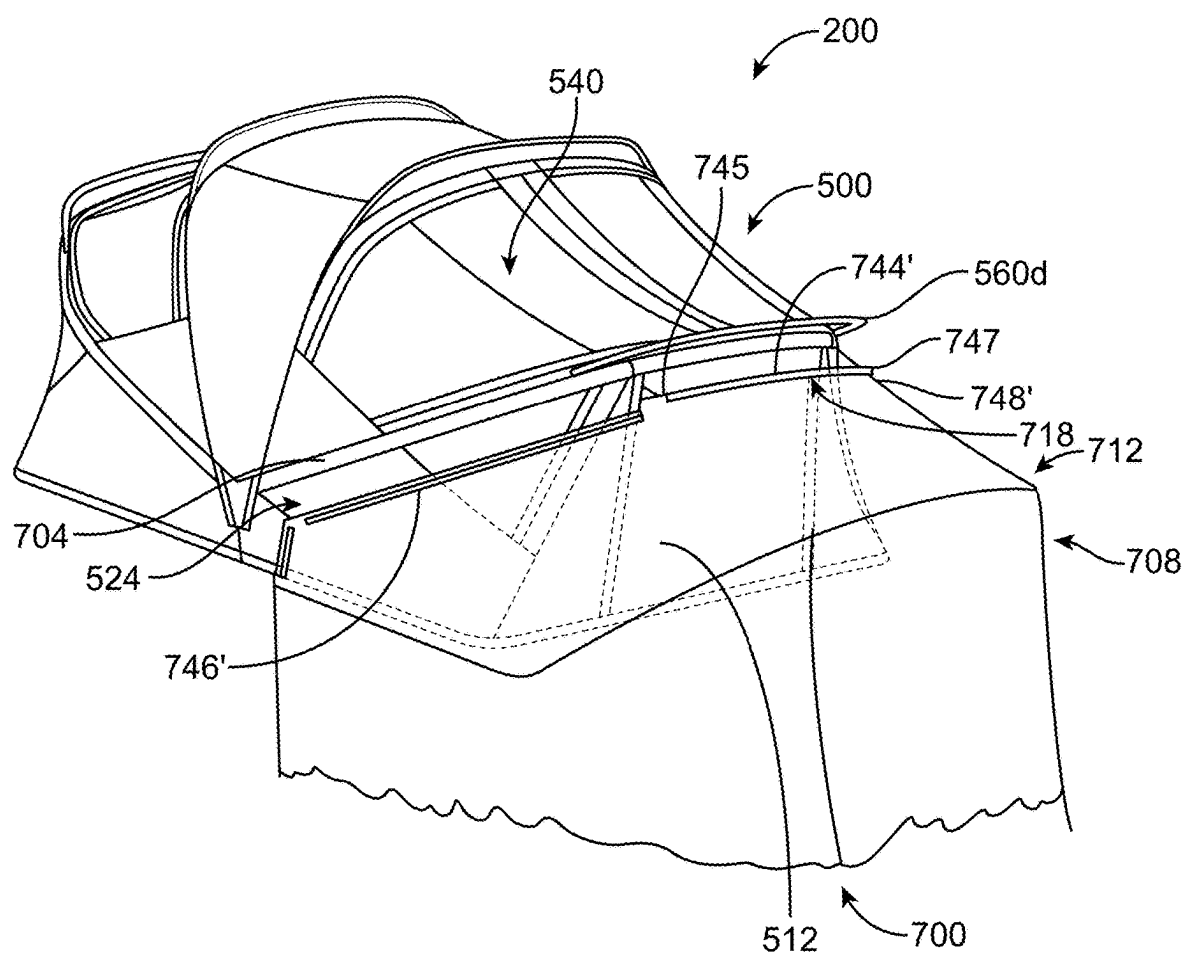


FIG. 15

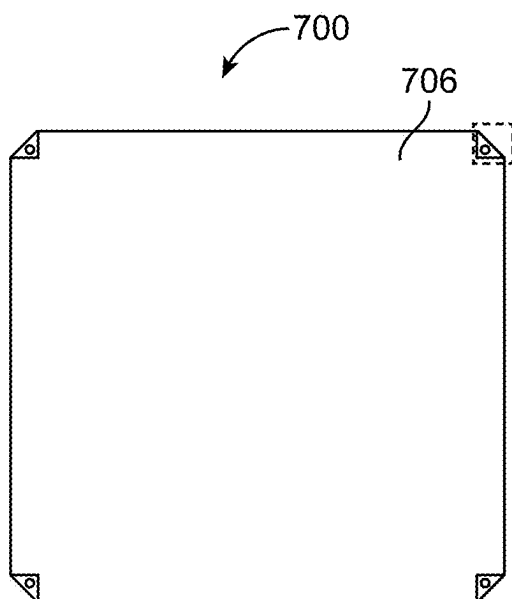


FIG. 16

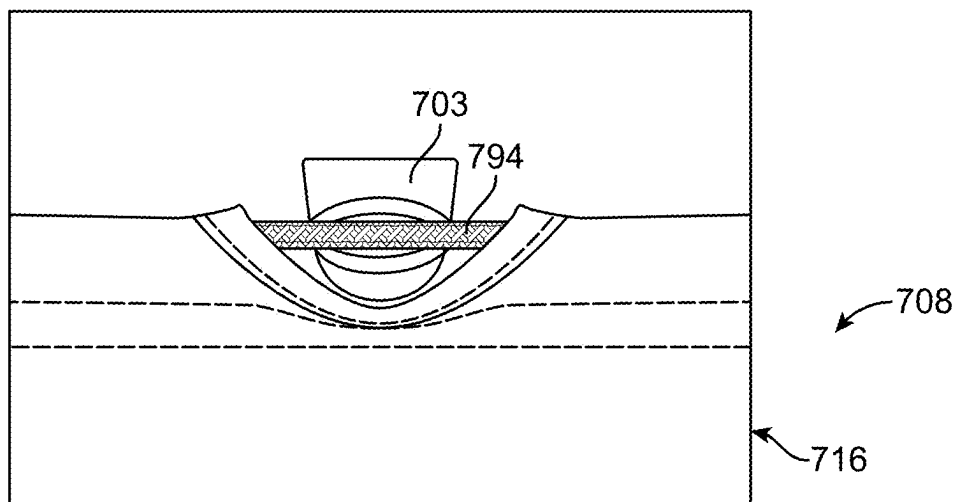


FIG. 17

ROOFTOP TENT ANNEX SYSTEM

BACKGROUND

Field

[0001] The present disclosure relates to tent apparatuses, systems, and methods, for example, adaptable rooftop tent apparatuses, systems, and methods having annex portions that can expand the interior volume of the rooftop tent apparatus.

Background

[0002] Current rooftop tents can utilize a plurality of tent poles attached to a base to support a main tent canopy and create an internal volume for a user on a vehicle rooftop. However, rooftop tents provide little additional space useful for camping. For example, the general shape of a rooftop of tent provides space to sit in or lie down, but may be limiting for standing room, for example, when changing clothes. Further, rooftop tents are generally limited with storage space and provide limited shelter to the neighboring campsite area from the elements, such as sun, rain, or snow.

BRIEF SUMMARY

[0003] Accordingly, there is a need to, for example, provide an annex that can be removably coupled to a rooftop tent to provide additional storage and functional space such that campers have a place to change clothes, sit or stand to shelter from the sun or inclement weather conditions, or provide additional storage for equipment while camping.

[0004] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the canopy. In some embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex is removably coupled to the base and the base canopy.

[0005] In some embodiments, the annex canopy is removably coupled to the base canopy and the annex frame assembly is removably coupled to the base. In some embodiments, the annex canopy is removably coupled to the base canopy with at least one zipper. In some embodiments, the annex frame assembly is coupled to a bottom side of the base. In some embodiments, the base canopy includes a door disposed on a first surface, and the annex canopy surrounds the door.

[0006] In some embodiments, the tent system further includes a ladder coupled to the base and extending between a floor and the base. In some embodiments, the annex completely surrounds the ladder.

[0007] In some embodiments, the annex canopy includes a lower opening portion. In some embodiments, the lower opening portion includes a plurality of side sections. In some embodiments, at least two of the plurality of side sections are configured to be in one of an opened configuration and a closed configuration.

[0008] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the base canopy. In some

embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex frame assembly is removably coupled to a bottom side of the base.

[0009] In some embodiments, the annex frame assembly is removably coupled to a base bracket disposed on the bottom side of the base. In some embodiments, the annex frame assembly includes a fastening member configured to slideably engage with an interior space formed in the base bracket. In some embodiments, the fastening member includes a T-bolt slideably disposed in the interior space and an actuator is threadably coupled with the T-bolt and configured to secure the annex frame assembly to the base bracket.

[0010] In some embodiments, the base bracket includes a first flange and a second flange spaced apart from the first flange to form a slot between the flanges. In some embodiments, the first and second flanges include engagement surfaces that have a shape corresponding to a connection portion of the annex frame assembly. In some embodiments, the engagement surfaces are configured to engage the connection portion of the annex frame assembly. In some embodiments, the connection portion of the annex frame assembly is cylindrical and the engagement surfaces of the first and second flanges are concave.

[0011] In some embodiments, the base bracket includes a stopper flange disposed in the interior space and configured to block movement of the fastening member. In some embodiments, the annex frame assembly does not extend through the base canopy.

[0012] In some embodiments, a tent system includes a base, a base canopy, a frame system, and an annex. In some embodiments, the base is configured to mount to a vehicle. In some embodiments, the frame system is coupled to the base and configured to support the base canopy. In some embodiments, the annex is configured to extend an interior space of the tent system in a lateral direction away from the base canopy and in a vertical direction toward the ground. In some embodiments, the annex includes an annex canopy and an annex frame assembly configured to support the annex canopy. In some embodiments, the annex canopy includes a height extendable portion configured to adjust a height of the annex canopy relative to the ground.

[0013] In some embodiments, the height extendable portion includes a first extendable section having a first section height, an upper coupling member disposed on an upper edge of the first extendable section, and a first lower coupling member disposed on a lower edge of the first extendable section. In some embodiments, the annex canopy is configured to be disposed in a first shortened configuration such that the first lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first section height.

[0014] In some embodiments, the upper coupling member and the lower coupling member include a zipper, a toggle and loop, a button and hole, or a hook-and-loop fastener. In some embodiments, the upper coupling member extends along a majority of the upper edge of the first extendable section, and the first lower coupling member extends along a majority of the lower edge of the first extendable section.

[0015] In some embodiments, the height extendable portion further includes a second extendable section having a second section height and extending from the lower edge of the first extendable section. In some embodiments, the first

lower coupling member is disposed on an upper edge of the second extendable section, and a second lower coupling member is disposed on a lower edge of the second extendable section.

[0016] In some embodiments, the annex canopy is configured to be disposed in a second shortened configuration such that the second lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first and second section heights. In some embodiments, the annex canopy is configured to be disposed in a second shortened configuration such that the second lower coupling member is coupled to the first lower coupling member and the height of the annex canopy is reduced by the second section height.

[0017] In some embodiments, the first section height is equal to the second section height. In some embodiments, the first section height is greater than the second section height.

[0018] In some embodiments, a tent annex system includes an annex canopy and an annex frame assembly. In some embodiments, the annex frame assembly includes a fastening member and a support pole. In some embodiments, the support pole is configured to support the annex canopy. In some embodiments, the fastening member of the annex frame is configured to removably couple to a base of a tent system. In some embodiments, the fastening member of the annex frame is configured to removably couple to a bottom side of a base of a tent system. In some embodiments, the annex canopy is configured to removably couple to a tent canopy of the tent system. In some embodiments, the annex system is configured to extend an interior space of the tent system in a lateral direction away from the vehicle and a vertical direction toward a floor.

[0019] Implementations of any of the techniques described above may include a system, a method, a process, a device, and/or an apparatus. The details of one or more implementations are set forth in the accompanying drawings and the description below. Other features will be apparent from the description and drawings, and from the claims.

[0020] Further features and advantages of the disclosure, as well as the structure and operation of various embodiments of the disclosure, are described in detail below with reference to the accompanying drawings. It is noted that the disclosure is not limited to the specific embodiments described herein. Such embodiments are presented herein for illustrative purposes only. Additional embodiments will be apparent to persons skilled in the relevant art(s) based on the teachings contained herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0021] The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate the embodiments and, together with the description, further serve to explain the principles and to enable a person skilled in the relevant art(s) to make and use the embodiments. Objects and advantages of illustrative, non-limiting embodiments will become more apparent by describing them in detail with reference to the attached drawings.

[0022] FIG. 1 illustrates an exploded perspective view of an annex and tent, according to an embodiment.

[0023] FIG. 2 illustrates a perspective view of an annex coupled with the tent, according to an embodiment.

[0024] FIG. 3 illustrates a side view of an annex and tent mounted on a vehicle, according to an embodiment.

[0025] FIG. 4 illustrates a bottom perspective view of an annex frame assembly coupled with a tent frame system, according to an embodiment.

[0026] FIG. 5 illustrates a detailed perspective view of an annex frame assembly coupled with a tent frame assembly, according to an embodiment.

[0027] FIG. 6 illustrates a cross-sectional view of the annex frame assembly along sectional line 6-6 shown in FIG. 5, according to an embodiment.

[0028] FIG. 7 illustrates a cross-sectional view of the annex frame assembly along sectional line 7-7 shown in FIG. 5, according to an embodiment.

[0029] FIG. 8 illustrates a side view of an annex and tent, with the annex in an extended configuration, according to an embodiment.

[0030] FIG. 9 illustrates a side view of an annex and tent, with the annex in a shortened configuration, according to an embodiment.

[0031] FIG. 10 illustrates an interior perspective view of an annex showing first, second, and third extendable sections, according to an embodiment.

[0032] FIG. 11 illustrates an interior perspective view of an annex in an intermediate shortened configuration, according to an embodiment.

[0033] FIG. 12 illustrates an exterior perspective view of an annex in the intermediate shortened configuration, according to an embodiment.

[0034] FIG. 13 illustrates a perspective view of a tent with an annex flap, according to an embodiment.

[0035] FIG. 14A illustrates a perspective view of a tent with a zipper configuration for coupling with the annex, according to an embodiment.

[0036] FIGS. 14B and 14C illustrate detailed views of the zipper configuration shown in FIG. 14, according to an embodiment.

[0037] FIG. 15 illustrates a perspective view of a tent with a zipper configuration for coupling with the annex, according to an embodiment.

[0038] FIG. 16 illustrates a top view of an annex floor, according to an embodiment.

[0039] FIG. 17 illustrates an attachment clip for coupling an annex to a base of a tent frame system, according to an embodiment.

[0040] The features and advantages of the embodiments will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0041] Embodiments of the present disclosure are described in detail with reference to embodiments thereof as illustrated in the accompanying drawings. References to “one embodiment,” “an embodiment,” “some embodiments,” etc., indicate that the embodiment(s) described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in

the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0042] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “on,” “upper,” “opposite” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or in operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0043] The term “about” or “substantially” or “approximately” as used herein indicates the value of a given quantity that can vary based on a particular technology. Based on the particular technology, the term “about” or “substantially” or “approximately” can indicate a value of a given quantity that varies within, for example, 1-15% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 5\%$, $\pm 10\%$, or $\pm 15\%$ of the value).

[0044] Before describing such embodiments in more detail, it is instructive to present example environments in which embodiments of the present disclosure can be implemented.

Example Tent

[0045] Tent system 200 can be configured to expand and collapse a frame system (e.g., frame system 400 shown in FIGS. 1-4) and a base canopy (e.g., canopy system 500 shown in FIGS. 1-4) coupled to the frame system. As shown in FIGS. 1-4, tent system 200 can include base system 300, frame system 400, canopy system 500, and/or rain fly system 600 (not shown). Base system 300 can be configured to mount to a vehicle and support frame system 400, canopy system 500, and/or rain fly system 600. Base system 300 can be coupled to frame system 400 and canopy system 500. Frame system 400 can be configured to support canopy system 500. Frame system 400 can be coupled to base system 300 and canopy system 500. Further details and implementations of the tent system 200 and associated components are described in U.S. Patent Application Ser. No. 63/300,502, filed on Jan. 18, 2022, the entire contents of which is incorporated by reference herein.

[0046] Base system 300 can be configured to mount to a vehicle (e.g., vehicle 342 shown in FIG. 3). As shown in FIGS. 1-4, base system 300 can include first base member 312 and second base member 314. First base member 312 can be configured to form a platform for tent system 200 atop a vehicle. First base member 312 can be further configured to receive a ladder. Second base member 314 can be configured to form a platform for tent system 200 atop a vehicle. Second base member 314 can be further configured to attach (e.g., be fixed) to a rack of a vehicle. As shown in FIG. 4, for example, first and second base members 312, 314 can be coupled to each other, for example, via hinge 318. Hinge 318 can be configured to open and close first and second base members 312, 314 to expand and contract tent system 200.

[0047] Frame system 400 can include first frame member 410, second frame member 420, third frame member 430, and fourth frame member 440 as shown, for example, in FIG. 4. In some embodiments, frame members 410, 420, 430, 440 can include aluminum, titanium, fiberglass, a

metal, a ceramic, a polymer, a combination thereof, and/or any other rigid material. In some embodiments, frame members 410, 420, 430, 440 can be configured to expand canopy system 500 to form an internal volume for a user. For example, in open configuration 10, frame members 410, 420, 430, 440 can contact canopy system 500 when a canopy is attached.

[0048] First frame member 410 can be configured to operate as a first knee connection of frame system 400. As shown in FIG. 4, first frame member 410 can include first support 412, first joint 414, and first leg 416. First support 412 can be configured to pivot about first connection 411 to first base member 312. First support 412 can be coupled to first base member 312 of base system 300. First support 412 can be coupled to first leg 416 via first joint 414. First joint 414 can be configured to pivot first leg 416 about distal end of first support 412, for example, similar to a human knee joint. First leg 416 can be configured to contact canopy system 500 (e.g., canopy sleeve 560 shown in FIGS. 1 and 2) and to expand canopy system 500 into an expanded position. Similar elements and features of second frame member 420, third frame member 430, and fourth frame member 440 are described in U.S. Patent Application Ser. No. 63/300,502, filed on Jan. 18, 2022, the entire contents of which is incorporated by reference herein.

[0049] Canopy system 500 can be configured to receive frame system 400 and provide weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.). Canopy system 500 can be further configured to support rain fly system 600. As shown in FIGS. 1-3, in some embodiments, canopy system 500 can include canopy 510 and canopy sleeves 560. Canopy 510 can be further configured to provide an internal volume for a user. In some embodiments, canopy 510 can include a polymer, a plastic, a thermoplastic, an elastomer, polyester (e.g., Mylar®), thermoplastic polyurethane (TPU), polyethylene (e.g., Dyneema®), ultra-high molecular weight polyethylene (UHMWPE) (e.g., Dyneema® composite fiber (DCF)), nylon, silicone, poly-cotton, canvas, a combination thereof, and/or any other water resistant material.

[0050] As shown in FIGS. 1-3, canopy 510 can include front surface 512 (e.g., a door), back surface 514, side surface 516, and top surface 518. Canopy 510 can further include first section 522, second section 524, third section 526, fourth section 530, and fifth section 540. As shown in FIG. 2, first section 522 (e.g., leftmost side) can be disposed between base system 300 and canopy sleeve 560a, second section 524 (e.g., rightmost side) can be disposed between base system 300 and canopy sleeve 560d, third section 526 (e.g., center) can be disposed between canopy sleeves 560b, 560c, fourth section 530 (e.g., left side) can be disposed between canopy sleeves 560a, 560b, and fifth section 540 (e.g., right side) can be disposed between canopy sleeves 560c, 560d.

[0051] In some embodiments, canopy 510 can include a plurality of canopy sleeves 560. For example, as shown in FIGS. 1-3, canopy 510 can include four canopy sleeves 560a, 560b, 560c, 560d. Canopy sleeves 560 can be configured to secure frame system 400 to canopy system 500. In some embodiments, canopy sleeves 560 and canopy 510 can be formed from the same material. For example, canopy sleeves 560 and canopy 510 formed from the same material can increase manufacturing efficiency and decrease cost. In some embodiments, for example, stitching and/or fabric thickness can be adjusted to form canopy sleeves 560 and

canopy 510 from the same material. In some embodiments, canopy sleeves 560 and canopy 510 can be integrally formed. In some embodiments, canopy sleeves 560 and canopy 510 can be formed from different materials. In some embodiments, canopy 510 and/or canopy sleeves 560 can include an elastic material.

Example Annex System

[0052] In some embodiments, tent system 200 can include an annex system 700 to expand an interior space of the tent system 200 and extend the interior space towards the ground. In some embodiments, annex system 700 couples to the base system 300 and/or canopy system 500. As shown in FIGS. 1-4 and 16, for example, annex system 700 can include annex floor 706, annex canopy 708, and/or annex frame assembly 710. Base system 300 can include base bracket 702 configured to couple with annex frame assembly 710 as shown, for example, in FIGS. 4-7. As will be described in further detail below, annex system 700 can couple with canopy system 500 to extend the weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) provided by the canopy system 500 to the ground.

[0053] In some embodiments, annex frame assembly 710 provides support to annex canopy 708 and couples to a base bracket 702 of base system 300 as shown, for example in FIGS. 1-7. In some embodiments, annex frame assembly 710 is slideably coupled with base system 300 at locations on both sides of tent system 200. Both annex frame assembly 710 and base system 300 include attachment features (e.g. base bracket 702 and fastening member 774) on a right side and a left side of tent system 200. Annex frame assembly 710 is configured so that a user can assemble one side of annex frame assembly 710 with one hand. For example, a user can slideably engage and couple a left side of annex frame assembly 710 with one hand, and slideably engage and couple a right side of annex frame assembly 710 with their other hand. Annex frame assembly 710 couples to an underside of base system 300, for example, so that the annex frame assembly does not extend through canopy 510 of canopy system 500.

[0054] In some embodiments, base bracket 702 is disposed on an underside of first base member 312 of base system 300. In the illustrative embodiment shown in FIGS. 4 and 5, base bracket 702 is fastened to first base member 312 with rivets, bolts, screws, or similar fastening devices. In some embodiments, base bracket 702 can be formed with first base member 312 such that base bracket 702 is integral with first base member 312. In some embodiments, base bracket can be disposed on an underside of second base member 314 such that an annex system can be assembled to the opposite side of tent system 200.

[0055] In some embodiments, base bracket 702 includes upper wall 720, first side wall 722, second side wall 724, first flange 726, and/or second flange 728. In some embodiments the flanges 726, 728 form an interior space 730 therebetween, as shown in FIGS. 5-7. In some embodiments, upper wall 720 is coupled with the underside of first base member 312, for example, at a distal end 313 of first base member 312. In some embodiments, first side wall 722 extends from an edge of upper wall 720 and is perpendicular to upper wall 720. In some embodiments, second side wall 724 is spaced apart from first side wall 722 and extends from an opposite edge of upper wall 720. In some embodiments, second side wall 724 is perpendicular to upper wall 720. In some

embodiments, first flange 726 extends toward second side wall 724 from (e.g., a distal end of) first side wall 722. Second flange 728 extends toward first side wall 722 from (e.g., a distal end of) second side wall 724. In some embodiments, first flange 726 and second flange 728 are vertically aligned on the same plane and spaced apart to form a slot or gap 731 therebetween. In some embodiments, first flange 726 includes first surface 733 that faces upper wall 720 and second surface 735 opposite first surface 733 that faces away from base bracket 702. In some embodiments, second flange 728 includes first surface 737 that faces upper wall 720 and second surface 739 opposite first surface 737 that faces away from base bracket 702.

[0056] In some embodiments, interior space 730 is formed between upper wall 720, first side wall 722, second side wall 724, first flange 726, and second flange 728 as shown, for example, in FIG. 6. Interior space 730 is configured to receive fastening member 774 of annex frame assembly 710. In some embodiments, base bracket 702 includes stopper 732, for example extending from upper wall 720, that extends into interior space 730 as shown in FIGS. 6 and 7. Stopper 732 is configured to block movement of annex frame assembly 710 beyond a predetermined assembly distance in base bracket 702. When the annex frame assembly 710 hits the stopper 732, it provides feedback to the user that that annex frame assembly 710 is in the proper location to secure it to the base bracket 710.

[0057] In some embodiments, annex frame assembly 710 includes one or more canopy support pole 770 to support annex canopy 708, and fastening member 774 to couple with base bracket 702 as shown, for example, in FIGS. 5-7. In some embodiments, canopy support pole 770 includes support portion 771 that provides support and shape to annex canopy 708, and connection portions 772 that couple with base brackets 702 on either side of base system 300. In some embodiments, connection portion 772 and support portion 771 can be separate components that are coupled together. In the illustrative embodiment in FIGS. 3 and 4, connection portion 772 and support portion 771 are integrally formed such that canopy support pole 770 is a single component. Connection portion 772 is orientated horizontally relative to the ground and locates beneath base bracket 702. In some embodiments, support portion 771 forms a C-shape and extends away from connection portion 772, for example, at approximately 45 degrees relative to and away from the ground and then spans across the width annex system 700.

[0058] In some embodiments, fastening member 774 couples with base bracket 702. In some embodiments, fastening member 774 includes T-bolt 776 and actuator 778 as shown, for example in FIGS. 5-7. In some embodiments, T-bolt 776 extends through connection portion 772 of canopy support pole 770 and couples with actuator 778. In some embodiments, T-bolt 776 is slideably disposed in interior space 730 of base bracket 702. In some embodiments, T-bolt 776 is supported by first surfaces 733, 737 of first and second flanges 726, 728. In some embodiments, first and second flanges are located between and engage with T-bolt 776 and connection portion 772. In the illustrative embodiment in FIG. 6, connection portion 772 has a circular cross-section and second surfaces 735, 739 have a concave shape to provide a greater surface area between the annex system 700 and the base system 300 to share load. In some embodiments, connection portion 772 and second surfaces 735, 739 can have corresponding shapes to share loads

therebetween that are non-circular. In the illustrative embodiment shown in FIGS. 6 and 7, actuator 778 is a knob and threadably engaged with T-bolt 776 such that when actuator 778 is rotated, a distance between T-bolt 776 and connection portion 772 is reduced such that T-bolt 776 and connection portion 772 couple with first and second flanges 726, 728. In some embodiments, actuator 778 can be a lever, handle, or other user interface and configured to couple T-bolt 776 and connection portion 772 and base bracket 702 together.

[0059] By way of example, during assembly, a user can align T-bolt 776 of the annex frame assembly 710 with the interior space 730 of base bracket 702 and slide T-bolt 776 into interior space 730. T-bolt 776 can slide along first surfaces 733, 737 until T-bolt 776 engages with stopper 732. Stopper 732 blocks further movement of annex frame assembly 710 so that T-bolt 776 does not extend beyond the end of base bracket 702. Stopper 732 also provides sufficient contact length between connection portion 772 of canopy support pole 770 and base bracket 702 so that any moments acting on annex system 700 are accommodated. Stopper 732 can also be positioned to maintain desired dimensions of the annex system 700 relative to the canopy system 500.

[0060] In some embodiments, annex canopy 708 couples to canopy system 500 so that annex canopy 708 surrounds front surface 512, including the front door, of canopy 510 as shown, for example, in FIGS. 1-3. In some embodiments, annex canopy 708 extends outwardly away from front surface 512 and is supported by canopy support pole 770 of annex frame assembly 710. In some embodiments, annex canopy 708 is disposed around the canopy support pole 770 and extends towards the ground, for example, so that it locates adjacent to annex floor 706. In some embodiments, annex canopy 708 has approximately vertical orientation as it extends towards the ground. In the illustrative embodiment shown in FIGS. 1-3, annex canopy 708 tapers outwardly as it extends towards the ground such that a horizontal cross-sectional area of the annex canopy 708 is greater near the bottom than at the top. Annex canopy 708 provides weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) and can be made from similar materials to canopy 510 as described above. Annex canopy 708 extends the interior space of the tent system 200 to provide additional space away from the main tent sleeping area, for example, for storage, getting changed, keeping wet items out of the main tent interior space, or to provide an area for pets to rest.

[0061] In some embodiments, annex canopy 708 includes annex upper portion 712 and lower opening portion 716 as shown, for example, in FIGS. 1-3. In some embodiments, annex upper portion 712 and lower opening portion 716 can be coupled together by stitching, gluing, or other suitable weather-proofing methods. In some embodiments, annex upper portion 712 and lower opening portion 716 can be formed from a continuous material.

[0062] In some embodiments, annex upper portion 712 couples with canopy 510 adjacent to canopy sleeve 560d, for example, between second section 524 and fifth section 540 as shown, for example, in FIGS. 1-3. In some embodiments, annex upper portion 712 surrounds front surface 512, and the front door of the tent, so that a user can remain protected from the weather as they exit the tent and enter the annex.

In some embodiments, annex upper portion 712 is supported by and coupled to canopy support pole 770 of annex frame assembly 710.

[0063] In some embodiments, annex upper portion 712 includes canopy connection portion 718, first side section 734, second side section 736, top section 738, front section 740, and annex canopy sleeve 742 as shown, for example, in FIGS. 1-3, 14, and 15. In some embodiments, first side 734 and second side 736 of annex upper portion 712 wrap around each side of second section 524 of canopy 510 to form side walls of the annex upper portion 712. In some embodiments, top section 738 is disposed between first side 734 and second side 736. In some embodiments, first side 734, second side 736, and top section 738 extend approximately parallel to fifth section 540 of canopy 510 and extend away from canopy 510 in a lateral direction to the side of vehicle 342. In some embodiments, top section 738 extends approximately horizontal to canopy support pole 770 of annex frame assembly 710 and then transitions into front section 740 with approximately vertical orientation.

[0064] In some embodiments, canopy connection portion 718 extends across the interface of first side 734, top section 738, and second side 736 with canopy 510, as shown, for example, in FIGS. 2, 3, 14A, and 15. In some embodiments, canopy connection portion 718 couples with canopy 510 along the interface between the section 524 and fifth section 540. In some embodiments, canopy connection portion 718 includes a connection mechanism that couples with a corresponding connection mechanism on canopy 510. For example, in some embodiments, canopy connection portion 718 can include a zipper that couples with a corresponding zipper on canopy 510. In some embodiments canopy connection portion 718 can couple with canopy 510 using a hook and loop fastener, where the hook portion is on one area and the loop portion is on a corresponding area. In some embodiments, canopy connection portion 718 can include buttons or buckles that couple with corresponding holes formed in or loops attached to canopy 510.

[0065] In some embodiments, canopy connection portion 718 can extend along a majority portion of the interface between annex system 700 and canopy system 500. In the illustrative embodiment in FIGS. 14A-C, canopy connection portion 718 includes first zipper 744 and second zipper 746. In some embodiments, first zipper 744 and second zipper 746 are spaced apart at a central location above the door and front surface 512. In some embodiments, each of the first zipper 744 and the second zipper 746 extend away from each other along the interface between annex system 700 and canopy system 500 and toward the hinge 318. In the illustrative embodiment in FIG. 15, canopy connection portion 718' includes first zipper 744', second zipper 746', and third zipper 748'. In some embodiments, first zipper 744' spans the width of the door and front surface 512 along the interface between annex system 700 and canopy system 500. In some embodiments, second zipper 746' is spaced apart from a first end 745 of first zipper 744' and extends away from first zipper 744' along the interface between annex system 700 and canopy system 500 and toward the hinge 318. In some embodiments, third zipper 748' is spaced apart from a second end 747, opposite the first end 745, of first zipper 744' and extends away from first zipper 744' along the interface between annex system 700 and canopy system 500 and toward the hinge 318 on the opposite side of tent system 200 to the second zipper 746'.

[0066] In some embodiments, canopy system 500 includes an annex flap 704 that covers and conceals canopy connection portion 718 as shown, for example, in FIG. 13. Annex flap 704 provides weather protection (e.g., rain, sun, heat, wind, snow, cold, etc.) to the zippers 744, 746 of canopy connection portion 718. For example, annex flap 704 can include a sheet of material that extends along the interface between the annex system 700 and the canopy system 500 and positioned above the canopy connection portion 718 so that the annex flap 704 covers canopy connection portion 718. Annex flap 704 can be made from similar materials to canopy 510 as described above.

[0067] In some embodiments, annex upper portion 712 can include annex canopy sleeve 742. Annex canopy sleeve 742 can be configured to secure annex canopy 708 to annex frame assembly 710. In some embodiments, annex canopy sleeve 742 can extend into an interior space of the annex system 700. In some embodiments, annex canopy sleeve 742 can extend above top section 738 of annex upper portion 712. In some embodiments, annex canopy sleeve 742 can include a sleeve recess (e.g., a pocket) configured to receive canopy support pole 770. In some embodiments, annex canopy sleeve 742 can include a zipper sleeve to receive and enclose at least a portion of a canopy support pole 770.

[0068] Annex canopy sleeve 742 can be formed from the same material and annex canopy 708 or canopy 510. For example, forming annex canopy sleeve 742, annex canopy 708, and canopy 510 from the same material can increase manufacturing efficiency and decrease cost. In some embodiments, for example, annex canopy sleeve 742 can be coupled to annex canopy 708 by stitching or annex canopy sleeve 742 can be integrally formed with annex canopy 708. In some embodiments, annex canopy sleeve 742 and annex canopy 708 can be formed from different materials.

[0069] In some embodiments, lower opening portion 716 provides access to the interior of the annex system 700 and tent system 200. In some embodiments, lower opening portion 716 includes first side panel 780, first door panel 782, first corner panel 784, front door panel 786, second corner panel 788, second door panel 790, and/or second side panel 792 as shown in FIGS. 1-3. In some embodiments, first side panel 780, first door panel 782, and first corner panel 784 extend from first side 734 of annex upper portion 712. In some embodiments, first corner panel 784, front door panel 786, and second corner panel 788 extend from front section 740 of annex upper portion 712. In some embodiments, second corner panel 788, second door panel 790, and second side panel 792 extend from second side 736 of annex upper portion 712. In some embodiments, panels of the lower opening portion 716 are coupled with sections of the annex upper portion 712 by stitching, gluing or other similar coupling methods. In some embodiments, panels of the lower opening portion 716 are formed from the same material as sections of the annex upper portion 712. In some embodiments, each of the panels 780, 782, 784, 786, 788, 790, 792 of lower opening portion 716 include anchor points that can be fixed to the ground to secure the annex system 700 to the ground and provide tension in the panels to create the desired shape of the annex canopy 708.

[0070] In some embodiments, each of first door panel 782, front door panel 786, and second door panel 790 can be opened to gain access tent system 200. For example, each of the door panels 782, 786, 790 can be rolled up and held in an open configuration, for example, by toggle and loop

configurations included in the top and base of each panel. In the illustrative embodiment in FIG. 12, each of the door panels 782, 786, 790 can include window meshes 791, 793 so that air can pass through the annex system 700 when the panel doors 782, 786, 790 are in a closed configuration. In some embodiments, lower opening portion 716 includes third side panel 795 adjacent and parallel with vehicle 342, and first, second, and third side panels 780, 792, 795 couple with the first base member 312 of base system 300. In some embodiments, first, second, and/or third side panels 780, 792, 795 include guide rope 794 along portions of an upper perimeter of the side panels 780, 792, 795. In some embodiments, base system 300 includes attachment clips 703, for example, disposed on the underside of first base member 312 as shown, for example, in FIG. 4. In some embodiments, the guide rope 794 portions couple with attachment clips 703 disposed on the underside of first base member 312 of base system 300 (example shown in FIG. 17).

[0071] In some embodiments, annex floor 706 is disposed on the ground under annex canopy 708 and provides a weather-proof surface for the user to sit or stand on, or store items on. In the illustrative embodiment in FIG. 16, annex floor 706 has square or rectangular shape that corresponds to the cross-sectional shape of annex canopy 708 at the interface with the ground. In some embodiments, annex floor 706 can be oversized and extend beyond the section panels of lower opening portion 716 to provide additional area at a campsite, for example, to sit on or place items on that is spaced apart from wet mud or similar. In some embodiments, annex floor 706 can include anchor points to anchor the annex floor 706 to the ground. In some embodiments, the anchor points in annex floor 706 can correspond to the anchor points in the lower opening portion 716 of annex canopy 708.

Example Height Extension Portion of Annex Canopy

[0072] In some embodiments, annex canopy 708 includes height extension portion 714 as shown, for example, in FIGS. 8-12. In some embodiments, height extension portion 714 is located between annex upper portion 712 and lower opening portion 716. In some embodiments, height extension portion 714 can be coupled with annex upper portion 712 and lower opening portion 716 by stitching, gluing, or other suitable weather proofing methods. In some embodiments, each of annex upper portion 712, height extension portion 714, and lower opening portion 716 can be formed from a continuous material.

[0073] Height extension portion 714 adjusts a height of annex canopy 708 between an extended height H_E (e.g., as shown in FIG. 8), a shortened height H_S (e.g., as shown in FIG. 9), and/or a plurality of discrete heights between extended height H_E and shortened height H_S . Height extension portion 714 adjusts a height of annex canopy 708 so that tent system 200 can be used on both a tall vehicle, such as an SUV, and a low vehicle, such as a sedan or station wagon. For example, if tent system 200 is coupled to a low vehicle, the annex canopy 708 can be adjusted to a shorter height via height extension system 714 to avoid excess annex canopy material laying on the ground. In contrast, if the tent system 200 is coupled to a tall vehicle, the annex canopy 708 can be adjusted to a taller height using the height extension portion 714 so that the annex canopy 708 extends all the way to the ground.

[0074] In some embodiments, height extension portion 714 includes first extendable section 750, second extendable section 752, and/or third extendable section 754. In the illustrative embodiment shown in FIG. 8, first extendable section 750 extends below annex upper portion 712, third extendable section 754 extends above lower opening portion 716, and second extendable section 752 locates below first extendable section 750 and above third extendable section 754. First extendable section 750 has height H1. Second extendable section 752 has height H2. Third extendable section 754 has height H3. In some embodiments, height H1, height H2, and height H3 are the same. In some embodiments, height H1, height H2, and height H3 are different.

[0075] In some embodiments, height extension portion 714 includes upper coupling member 756, first lower coupling member 758, second lower coupling member 760, and/or third lower coupling member 762 as shown, for example, in FIGS. 8 and 10. In the illustrative embodiment in FIG. 10, the coupling members 756, 758, 760, 762 are zippers and extend around the perimeter of height extendable portion 714. In some embodiments, each of coupling members 756, 758, 760, 762 can be toggle and hoops, button and holes, hook and loop, or any other suitable coupling configuration. In some embodiments, upper coupling member 756 is coupled to an upper edge of first extendable section 750. In some embodiments, first lower coupling member 758 is coupled at the interface between a lower edge of first extendable section 750 and an upper edge of second extendable section 752. In some embodiments, second lower coupling member 760 is coupled at the interface between a lower edge of second extendable section 752 and an upper edge of third extendable section 754. In some embodiments, third lower coupling member 762 is coupled at a lower edge of third extendable section 754.

[0076] The height of annex canopy 708 is extended height H_E when none of the coupling members 756, 758, 760, 762 are coupled to each other. The height of annex canopy 708 can be adjusted shorter than the extended height H_E by coupling different coupling members 756, 758, 760, 762 in different combinations. For example, the height of the annex canopy 708 can be adjust to shortened height H_S by coupling third lower coupling member 762 with upper coupling member 756. In the shortened height H_S configuration, the lower edge of the third extendable section 754 couples with the upper edge of the first extendable section 750 and the height of the height extension portion 714 of the annex canopy 708 is removed from the total height of the annex canopy 708.

[0077] In a first example configuration, illustratively shown in FIG. 10, second lower coupling member 760 can be coupled with first lower coupling member 758 to reduce the height of annex canopy 708 by height H2. In this first example configuration, second extendable section 752 is folded over so that the upper and lower edges of second extendable section 752 become connected.

[0078] In a second example configuration, third lower coupling member 762 can be coupled with first lower coupling member 758 to reduce the height of annex canopy 708 by the combined height of height H2 plus height H3. In this second example configuration, second extendable section 752 and third extendable section 754 are folded over so that the upper edge of second extendable section 752 is coupled to the lower edge of third extendable section 754.

[0079] In an illustrative example shown in FIGS. 11 and 12, second lower coupling member 760 couples with upper coupling member 756. In the illustrative configuration, the height of annex canopy 708 has been shortened by the combined height of height H1 plus height H2. In the illustrative configuration, only the third extendable section 754 is visible from the interior of the annex system 700. In the illustrative configuration, first and second extendable sections 750, 752 are folded over on the outside of annex system 700.

[0080] In some embodiments, height extension portion 714 can include more than three extendable portions to provide additional adjustment in the height of the annex canopy 708. In some embodiments, height extension portion 714 can include less than three extendable portions to provide additional adjustment in the height of the annex canopy 708. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2300 mm to about 1500 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2150 mm to about 1700 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2100 mm to about 1850 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2050 mm to about 1900 mm. In some embodiments, the height extension portion can adjust an annex canopy from a height of about 2050 mm to about 1950 mm.

[0081] It is to be appreciated that the Detailed Description section, and not the Brief Summary and Abstract sections, is intended to be used to interpret the claims. The Summary and Abstract sections may set forth one or more but not all embodiments of the support assembly system and apparatus, and thus, are not intended to limit the present embodiments and the appended claims.

[0082] The present disclosure has been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

[0083] The foregoing description of the specific embodiments will so fully reveal the general nature of the disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

[0084] The breadth and scope of the present disclosure should not be limited by any of the above-described embodiments, but should be defined only in accordance with the following claims and their equivalents.

1. A tent system, comprising:
a base configured to mount to a vehicle;
a base canopy;
a frame system coupled to the base and configured to support the base canopy; and
an annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex is removably coupled to the base and the base canopy.
2. The tent system of claim 1, wherein the annex canopy is removably coupled to the base canopy and the annex frame assembly is removably coupled to the base.
3. The tent system of claim 1, wherein the annex canopy is removably coupled to the base canopy with at least one zipper.
4. The tent system of claim 1, wherein the annex frame assembly is coupled to a bottom side of the base.
5. The tent system of claim 1, wherein the base canopy comprises a door disposed on a first surface, and the annex canopy surrounds the door.
6. The tent system of claim 1, further comprising a ladder coupled to the base and extending between the floor and the base, wherein the annex completely surrounds the ladder.
7. The tent system of claim 1, wherein the annex canopy includes a lower opening portion, the lower opening portion having a plurality of side sections and at least two of the plurality of side sections are configured to be in one of an opened configuration and a closed configuration.
8. A tent system, comprising:
a base configured to mount to a vehicle;
a base canopy;
a frame system coupled to the base and configured to support the base canopy; and,
an annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex frame assembly is removably coupled to a bottom side of the base.
9. The tent system of claim 8, wherein the annex frame assembly is removably coupled to a base bracket disposed on the bottom side of the base.
10. The tent system of claim 9, wherein the annex frame assembly comprises a fastening member configured to slideably engage with an interior space formed in the base bracket.
11. The tent system of claim 10, wherein the fastening member comprises a T-bolt slideably disposed in the interior space and an actuator threadably coupled with the T-bolt and configured to secure the annex frame assembly to the base bracket.
12. The tent system of claim 9, wherein the base bracket includes a first flange and a second flange spaced apart from the first flange to form a slot therebetween, the first and second flanges comprise engagement surfaces that have a

shape corresponding to a connection portion of the annex frame assembly and are configured to engage the connection portion of the annex frame assembly.

13. The tent system of claim 12, wherein the connection portion of the annex frame assembly is cylindrical and the engagement surfaces of the first and second flanges are concave.

14. The tent system of claim 10, wherein the base bracket includes a stopper disposed in the interior space and configured to block movement of the fastening member.

15. The tent system of claim 8, wherein the annex frame assembly does not extend through the base canopy.

16. A tent system, comprising:

a base configured to mount to a vehicle;
a base canopy;

a frame system coupled to the base and configured to support the base canopy; and,

an annex configured to extend an interior space of the tent system in a lateral direction away from the base canopy and in a vertical direction toward the ground, the annex comprising an annex canopy and an annex frame assembly configured to support the annex canopy, wherein the annex canopy includes a height extendable portion configured to adjust a height of the annex canopy relative to the ground.

17. The tent system of claim 16, wherein the height extendable portion comprises a first extendable section having a first section height, an upper coupling member disposed on an upper edge of the first extendable section, and a first lower coupling member disposed on a lower edge of the first extendable section, wherein the annex canopy is configured to be disposed in a first shortened configuration such that the first lower coupling member is coupled to the upper coupling member and the height of the annex canopy is reduced by the first section height.

18. The tent system of claim 17, wherein the upper coupling member and the lower coupling member comprises a zipper, a toggle and loop, a button and hole, or a hook-and-loop fastener.

19. The tent system of claim 17, wherein the upper coupling member extends along a majority of the upper edge of the first extendable section, and the first lower coupling member extends along a majority of the lower edge of the first extendable section.

20. The tent system of claim 17, wherein the height extendable portion further comprises a second extendable section having a second section height and extending from the lower edge of the first extendable section, the first lower coupling member disposed on an upper edge of the second extendable section, and a second lower coupling member disposed on a lower edge of the second extendable section.

21-26. (canceled)

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